

repairs more efficiently, reducing system downtime. For example, maintenance personnel can access the virtual twin to identify the exact components, understand their dimensions, and review historical maintenance records. This enables more efficient and targeted maintenance activities, minimizing downtime and extending the asset's useful life. Similarly, they can utilize the virtual twin to understand the system's physical layout, access metadata for part identification, and troubleshoot issues without direct access to the physical asset.

2.2.4. Obsolescence management

Obsolescence management is critical for ensuring the continuous operational readiness, sustainability, and effectiveness of certain systems over their extended service lives, which can often span several decades. Effective obsolescence management requires a proactive and holistic approach, integrating various strategies throughout the system's life cycle to ensure operational readiness, reduce risks, and manage costs associated with maintaining and supporting these systems over time. The use of digital models combined with some emergent manufacturing technologies, provides new tools for obsolescence management.

Geometric information embedded in digital twins makes it easier to reverse engineer certain parts and manage obsolescence, as they are exact replicas of their physical twins, enabling their manufacturing. Integrating CAD and Computer-Aided Manufacturing (CAM) systems streamlines product design and manufacturing processes, which is especially helpful for modernization and dealing with obsolete parts. This link can be further reinforced by novel manufacturing processes, such as additive manufacturing (also known as 3D printing), as the link between design and manufacturing can be entirely established as a digital thread.

2.3. Digitally transforming system retirement

Relative to other life cycle phases, system retirement is the least developed with respect to digital transformation. For some situations, it may be advantageous to preserve end use data from retired systems and components that have reached end of life, such as performance data, as the same component in the retiring system may continue to be used in other similar applications. As an example, if you retire a single aircraft, a certain actuator may still have usable life for use in other aircraft. Performance data may also be used for

future simulations to influence future redesigns. The digital thread can also help in better identifying and dealing with components with special considerations for disposal, such as hazardous materials or classified hardware and software.

Furthermore, the retirement stage of a given system for one organization could be, simultaneously, the start of an acquisition program for another organization (e.g., when an organization acquires a second-hand aircraft from another organization). In this case, leveraging or even accepting the digital twin and digital thread developed with the system can be challenging for the organization who is receiving the assets, given that their digital transformation process may not be mature enough.

By leveraging digital twins and digital threads in the retirement stage, organizations can enhance sustainability, compliance, and efficiency in the decommissioning process. These technologies contribute to responsible and environmentally friendly practices while providing valuable insights for ongoing improvement in the development and life cycle management processes.

2.4. Examples of novel capabilities digitally transforming the system life cycle

2.4.1. Cloud-enable collaborative model development

Collaborative modelling tools facilitate cross-functional collaboration, which may be integrated with Product Life cycle Management (PLM) CAD suites to allow engineers, designers, and other stakeholders to work together seamlessly. Changes made in one tool can be communicated to the other, promoting communication and alignment between different teams.

The use of collaborative model development technology supports formalizing model planning, development, integration, curation, and using models for engineering activities and decision-making across the life cycle. As opposed to formal document-based approaches in legacy systems engineering practices, model-centric organizations leverage collaborative environments where teams can define and plan the creation of models to support engineering activities, ensuring a structured and auditable approach. This technology facilitates the formal development of models by providing tools incorporating various techniques and algorithms, integrating data from different sources, and

curating models by refining and optimizing them over time. Significant productivity gains can be made with emphasis on the model-based definition and evolution from the legacy document-based approaches.

2.4.2. Data aggregation and integrated information systems

For most organizations, there is a significant labor burden associated with manual processes and disparate systems used to collect and use data. The modern information system enabled by digital threads may connect various disciplines and stages of the asset life cycle. Virtual product and process models, represented by digital twins and the digital thread, can enable analysis, performance optimization, decision-making, operations, by integrating data and information for downstream users.

Digital twin technology allows real-time (or near-real time) virtual representations of the current asset state, in which databases, repositories, and system models from multiple disciplines may be integrated. A well-architected digital thread can provide seamless integration and accessibility of relevant data across the asset's entire life cycle, from design to operation and sustainment to enable functionality in the digital twin.

2.4.3. Sensors and IoT

The emergence of novel capabilities in sensor technology and the implementation of the Internet of Things (IoT) have revolutionized the way in which digital twins are developed, offering a seamless integration of the physical and digital domains. In parallel, advances in microfabrication and nanotechnology have led to the creation of highly sensitive, compact, and energy-efficient sensors capable of detecting a wide array of physical phenomena, necessary to develop more realistic digital twins.

Concurrently, innovations in IoT technology, including enhanced connectivity options, robust data processing, and cloud computing, facilitate the reliable transmission and analysis of vast amounts of data collected from these sensors. This IoT infrastructure enables the real-time synchronization of physical assets with their digital counterparts, creating dynamic, virtual models that accurately reflect the physical world.

2.4.4. Cloud computing

The emergence of cloud computing has revolutionized system deployments. Cloud platforms provide on-demand access to resources, enabling organizations to easily scale their systems and deploy them globally. Cloud-based deployment models, such as Infrastructure as a Service (IaaS) and Platform as a Service (PaaS), simplify the deployment process for organizations by abstracting underlying infrastructure concerns. Specifically, they reduce the need to heavily invest in physical servers and data centers.

Infrastructure as a Service (IaaS) is a category of cloud computing services that provides virtualized computing resources over the internet. In an IaaS model, users can rent or lease various infrastructure components, such as virtual machines, storage, and networking, instead of investing in and maintaining their own physical hardware.

Platform as a Service (PaaS) is a cloud computing service model that provides a platform allowing customers to develop, run, and manage software applications without the complexity of building and maintaining the underlying infrastructure. In a PaaS model, the cloud provider delivers a comprehensive and integrated platform that includes development tools, runtime environments, and other services necessary for building, deploying, and scaling applications.

3. CONSIDERATIONS FOR PURSUING DIGITAL TRANSFORMATION ACROSS THE LIFE CYCLE

3.1. Overview

Digital transformation across the life cycle demands a comprehensive approach to technical data management strategy. This strategy should guide the acquisition, management, and maintenance of the technical data and software necessary to support a system from inception to retirement. Considerations such as safeguarding intellectual property and fostering competition should be central to the strategy. By securing the required data and rights, organizations can enhance system design understanding, optimize operations across varied environments, and unlock potential cost efficiencies in acquisition and sustainment.

The effectiveness of digital models and digital twin technology is deeply linked to their accessibility and usability by

operational users. These stakeholders depend on intuitive tools that allow for the realistic simulation of digital twins, alongside seamless access to comprehensive logistics and technical data. This data should be open, interoperable, adaptable, and accessible across cloud and edge computing environments, necessitating a collaborative effort that integrates the insights of both technical teams and operational users right from the design phase.

Ideally, a single digital twin would suffice for all life cycle phases of a system, incorporating all necessary information, models, and behaviors. However, embedding operational and sustainment capabilities effectively in the digital twin requires attention during the concept, development, and production stages. The transition towards a unified digital twin demands a gradual integration of operational and product support models. Until such comprehensive integration is achieved, distinctions among design, operational, and sustainment digital twins remain essential. Without careful implementation, there is a risk of creating a digital twin that serves well for design and production but falls short in operations and sustainment due to an inability to replicate its physical twin's behaviors accurately. Thus, a true digital twin must effectively embody both design and operational and support functionalities.

The development of a holistic digital twin, equipped with the necessary capabilities and functionality to support desired use cases, can be regarded as a standalone software development project. This project should proceed in tandem with the development of the System of Interest, highlighting the need for a multidisciplinary approach. Achieving the goals of digital transformation necessitates a holistic view that encompasses people, processes, technology, data, and strategic objectives, all while maintaining a focus on establishing a durable digital thread that interconnects all elements of the life cycle.

Some guidance to implement and/or adopt digital engineering for later phases of the system life cycle is provided in the following sections.

3.2. When in doubt, start with a pilot

There is significant upfront work required in preparation for downstream use of digital twin and digital thread, including in the system concept and development phases. It can be overwhelming to figure out where to start, which use cases to prioritize, and what resources will be needed to mature the capability.

For many organizations, digital transformation efforts begin with a pilot project, followed by what is typically a slow, challenging transition from successful pilot projects to scaled operations [8]. Successful transformation at scale requires thoughtful planning and coordination to holistically implement the new technology within the context of the organization. The transformation should be aligned to strategic goals and commitment to from leadership to critically evaluate legacy processes, frameworks, and architectures and make appropriate investments in driving digital transformation objectives. It also involves leadership commitment to empowering innovative teams, and fostering a culture that is open to experimentation and capable of learning from failure.

3.3. Be cognizant of technical data rights

Organizations must plan in advance for the technical data rights needed to be acquired if the system is designed by an external entity. If this work is not done upfront, many digital transformation opportunities may not be achievable due to proprietary or incompatible formats, inaccessible data, or cost of rework for the system to produce desired data. In all projects, sponsors should anticipate technical data required for life cycle activities and establish rights or options to purchase data in contracts and service level agreements [9], as well as consult with experts regarding the appropriate data and technology standards to invoke in contract specifications.

3.4. Leverage standardization.

It is essential to adopt semantically rich, open, and accessible data standards, which facilitate interoperability, data linkage, and contextualization that all stakeholders can use and build upon for their needs. Emphasis on open data standards cannot be overstated. In order to have comprehensive data integration and offer a holistic view of the asset's history, current state, and simulate future states, the digital thread must support integration of data from different sources, such as design data, manufacturing data, sensor data, maintenance logs, and operational data.

Various standards development organizations (SDOs) have developed specifications to address interoperability in digital twins [10]. Some examples are provided in Table 1. Furthermore, the domain of digital twins has witnessed a growing expansion of open-source activities, particularly in the development of digital twin platforms and data management. These open-source initiatives contribute to collaboration, innovation, and the wider adoption of digital twins.

Standards Development Organization (SDO)	Scope
ISO/TC 184	Establishes industrial data standards across different domains, including manufacturing, industrial automation, and information systems to ensure the compatibility and interoperability of digital twins in the smart factory field.
IEEE P3144 Digital Twin Working Group	The Standard for Digital Twin Maturity Model and Assessment Methodology in Industry defines a digital twin maturity model for industry, including digital twin capability domains and corresponding subdomains. This standard also defines assessment methodologies, including assessment content, assessment processes, and assessment maturity levels.
The 3rd Generation Partnership Project (3GPP)	Focused on developing standards for 5G networks, which offer the high-speed and reliable communication capabilities required for digital twins.
The Open Geospatial Consortium (OGC)	Manages geospatial information standards, which are crucial for digital twins in smart cities and other domains.
IEC TC65	Focuses on interoperability standards, specifically in the smart factory context. Its efforts help to harmonize communication and data exchange between various components of digital twins within the manufacturing domain.
oneM2M	A global initiative that standardizes service layer IoT platforms, providing common service functions that are essential for the effective operation of digital twins.

Table 1. Standardization in digital twin technology.

3.5. Connect traceability and configuration management across the life cycle

Organizations must plan for traceability to ensure data is properly managed and easy to find, and that data flows and interfaces are maintained in operations and sustainment for the delivered systems. This may be difficult to invoke contractually even in a highly regulated industry, particularly in complex systems designed with disparate development processes. Furthermore, technical and logistics information of a system may reside and be maintained in disparate systems. Linking the Logistics Supportability Analysis Record (LSAR) databases and underlying data sets requires a blueprint for establishing and maintaining these connections, especially in distributed and disconnected environments.

3.6. Intentionally enable automation and analytics

An organization undergoing digital transformation may seek sensor technology and monitoring datasets to generate operational profiles or support development of behavior models. Monitoring technology may already be present in the system, displaying the condition of the asset or its behavior. Modelling and simulation of the remote asset or fleet is made possible by combining real-time data with historical and contextual information. In practice, this requires significant data capture and data usability, including careful planning to maintain linkages between authoritative data sources and robust methods to maintain links between authoritative data and automated methods to ingest multiple sources and formats in secure environments. However, lack of a pre-existing process foundation and the need to streamline or transform existing processes are significant barriers for organizations to adopt automation [11].

3.7. Actively manage knowledge to support operations & sustainment

When introducing new technology in a production system, the workforce usually undergoes training for a smooth system deployment. This generally requires unifying relationships, processes, and data across the production system operations, many of which cross organizational boundaries. Digital threads can facilitate the transfer of knowledge and lessons learned as an organization integrates and transitions from one asset to another. Historical data, maintenance actions, and failure analysis from similar assets can be shared and used to improve predictive maintenance strategies. Furthermore, digital threads can also enable collaboration and knowledge sharing. Authorized personnel, such as operators, engineers, or subject matter experts, can access the digital thread to share insights and knowledge, exchange information, maintain data libraries, or provide support. This collaborative environment enhances problem-solving capabilities and facilitates decision-making processes.

3.8. Implementation impacts on life cycle

Developing digital twins for complex systems presents multifaceted challenges, requiring a strategic approach from the outset. The development of digital twins generally begins with a focus on design and production, but this approach must evolve so that the digital twin can be used throughout the entire system life cycle, from deployment through operations, maintenance, and eventual retirement.

The inherent value of digital twins lies in their ability to enhance the availability, accessibility, and accuracy of information. This is particularly true when data can be contextualized within the operational environment of users. Such capabilities, conceived at early life cycle stages, are instrumental in reaping long-term benefits. They facilitate not just a smoother integration of new technologies but also support the workforce as it navigates shifting paradigms. Adjusting traditional program and project management practices and making upfront investments in a robust digital infrastructure are essential steps in this direction.

A critical phase in this progression involves augmenting the digital twin traditionally employed for design and production with functionalities that are necessary to sustain the system once it has transitioned into operations. In line with systems engineering principles, engaging end-users, operators, and maintainers from the beginning in drafting operational concepts, defining scenarios, and setting requirements is key to success.

4. CONCLUSIONS

Digital engineering, underpinned by Product Life Cycle Management (PLM) and Model-Based Systems Engineering (MBSE) frameworks, plays a crucial role in the deployment, operations and sustainment, and retirement phases of system life cycle management. These methodologies facilitate the creation of digital twins and digital threads, which serve as dynamic virtual models mirroring real-world assets throughout their service life. Such digital foundations enable continuous evolution and improvement of systems by providing detailed blueprints for sustainment activities, such as maintenance. This comprehensive approach ensures that systems not only meet initial requirements but also adapt to future needs, thereby extending their utility and enhancing overall performance.

The implementation of digital engineering principles faces challenges, including organizational resistance to change, outdated infrastructure, and skill gaps. Overcoming these obstacles requires a focused strategy that highlights the transformation's unique benefits, such as improved efficiency and streamlined operations. Additionally, the rapid pace of technological advancement necessitates a flexible and collaborative digital transformation strategy, allowing organizations to explore innovative solutions and adapt to new challenges. Successful digital transformation also involves treating data as a strategic asset, establishing strong data governance, and ensuring data quality and security. By prioritizing these elements, organizations can maximize the benefits of digital engineering, leading to more reliable, cost-effective, and high-performing systems throughout their life cycle.

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BIOGRAPHIES

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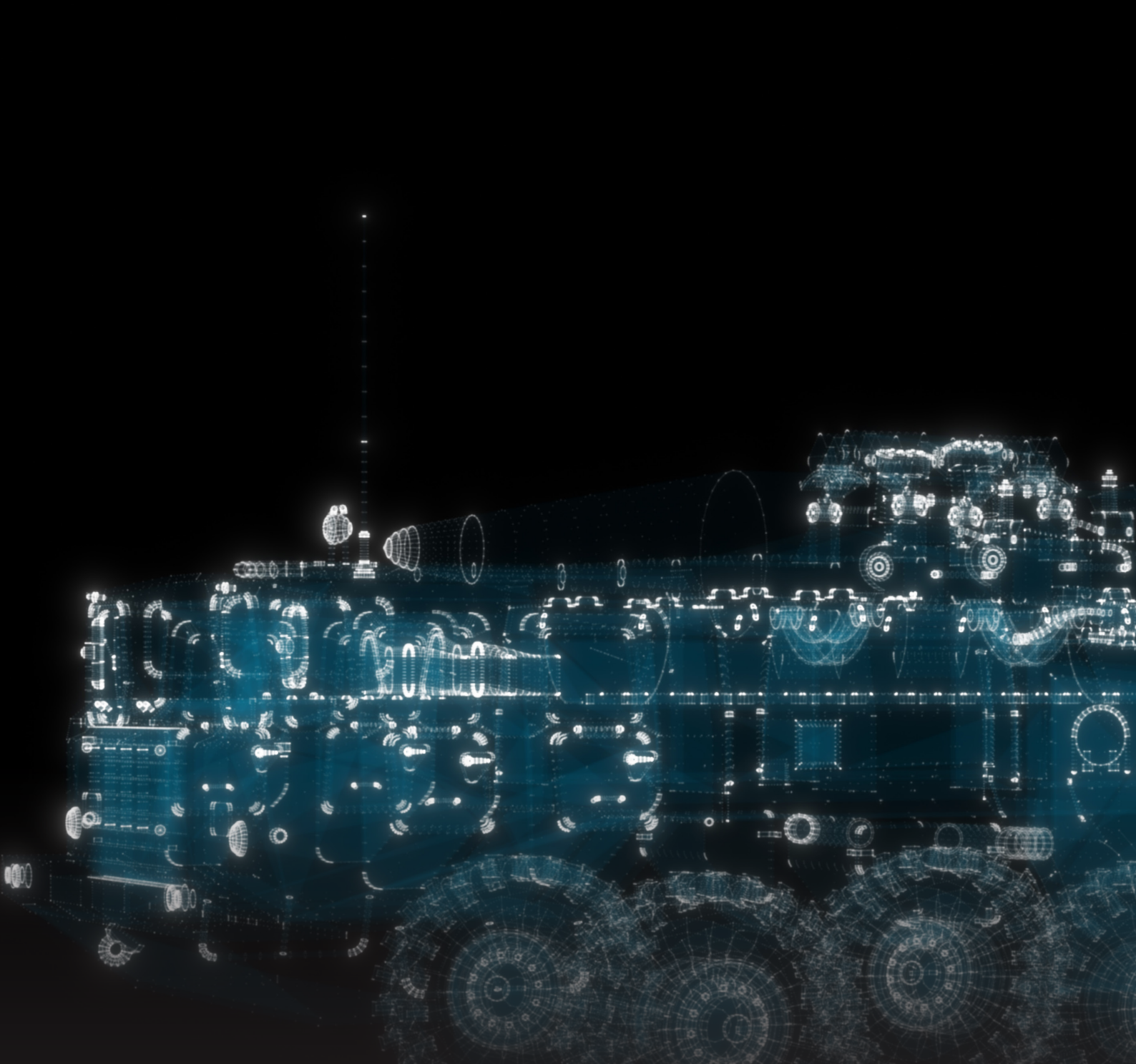
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"Systems engineering requires knowledge and skill. It is impossible to acquire systems engineering skill in a short course or workshop! [...] It is impossible to learn systems engineering from a systems engineering tool."

A.W. Wymore

The previous six chapters are evidence of the profound transformation that the landscape of systems engineering is facing. Over the past three decades, we have witnessed a remarkable journey of growth, innovation, and maturation of the field. We have seen a considerable expansion of knowledge, the embracing of new methodologies, and the breaking of traditional boundaries that once confined our understanding and application of systems engineering principles. The field has changed in ways that were unimaginable at the time of Isdefe's "blue books" publication.

The conclusion of this first monograph on modern systems engineering marks a new chapter in the dissemination of the discipline in Spain. This monograph marks the beginning of the new "blue books" series. It is both a testament to the progress made and a compass for the future. It encapsulates some key topics of the current state of practice in systems engineering, highlighting nascent areas and innovative practices that are shaping the future of the field. The diversity of topics covered—from the application of model-based systems engineering (MBSE) to the integration of artificial intelligence and digital twins in system lifecycle management—illustrates the multifaceted nature of systems engineering and its significance in a rapidly changing world. In the future, we intend to release additional monographs in the series to dive deeper into each of these topics. We also conceive the dissemination of these advances as opportunities for Isdefe engineers to succeed in the challenges they face, in the technically advanced programs of our society, particularly in these times of growing insecurity and uncertainty.

It is with a sense of pride and optimism that we release this monograph into the world. Our aim is for it to serve as a starting point for practitioners, guiding them through the complexities of modern engineering challenges and inspiring them to push the boundaries of what is possible in systems engineering. We extend our deepest gratitude to all who have contributed to this work, and to the readers, who have the challenging task of adopting and evolving the modern systems engineering practices we have presented here.

Let this epilogue not signify the end, but rather the commencement of a journey towards new frontiers in systems engineering and its wider adoption. May the insights contained within these pages inspire you, driving forward the development of engineered systems that are not only fit for purpose but also resilient and sustainable in the face of future challenges. Together, we stand on the threshold of a new era, ready to explore, innovate, and shape the future of systems engineering for the betterment of society and the world at large.

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